



ROBOTICS GUILD

Building Robots for Fun - Simple Bots to Complex Autonomous Systems



GUILD PHILOSOPHY

Building robots for the joy of creation and discovery

From Simple to Complex:

Minimal sensors and drives to fully autonomous mobile systems

All Scales Welcome:

Tiny desktop bots to large mobile platforms

Learn by Building:

Hands-on projects, experimentation, and iteration

Share and Collaborate:

Community support, group builds, knowledge exchange

Multiple processors • Complex sensing • Autonomous movement • AI integration



ROBOT TYPES AND PLATFORMS

Simple Bots

Line followers
Obstacle avoiders
Light seekers
Remote controlled
Educational kits

Mobile Robots

Wheeled platforms
Tracked vehicles
Differential drive
Omnidirectional
Outdoor rovers

Manipulators

Robotic arms
Grippers and end effectors
Multi-axis systems
Pick and place
Delta robots

Legged Robots

Bipeds
Quadrupeds
Hexapods
Walking gaits
Balance control

Aerial Robots

Quadcopters
Fixed-wing drones
FPV racing drones
Autonomous flight
Camera platforms

Specialized

Underwater ROVs
Combat robots
Art and performance
Agricultural bots
Social robots



SENSORS AND PERCEPTION

Vision

Cameras (RGB, thermal)
Depth sensors (stereo, ToF)
LiDAR
Computer vision
Object detection
Visual SLAM

Distance and Proximity

Ultrasonic sensors
Infrared distance
Laser rangefinders
Bump sensors
Capacitive touch
Cliff detection

Motion and Orientation

IMU (accelerometer, gyro)
Magnetometer (compass)
Encoders (wheel, motor)
GPS/GNSS
Optical flow
Odometry

Environmental

Temperature
Humidity
Air quality
Light levels
Sound (microphones)
Gas detection

Force and Touch

Force sensors
Torque sensors
Pressure sensors
Tactile arrays
Strain gauges
Load cells

Location Aware

GPS positioning
Indoor positioning
SLAM (simultaneous localization)
Beacon tracking
Map building
Path planning



ACTUATORS AND CONTROL

Motors

DC motors
Servo motors
Stepper motors
Brushless motors
Linear actuators

Motor Control

H-bridge drivers
PWM control
PID controllers
ESCs (electronic speed)
Motor encoders

Other Actuators

Pneumatic cylinders
Hydraulic systems
Solenoids
Speakers and buzzers
LEDs and displays

Power Systems

LiPo batteries
Battery management
Voltage regulators
Power distribution
Solar charging

Mechanical Systems

Gears and gearboxes
Belts and pulleys
Chains and sprockets
Linkages and joints
Bearings and bushings

Grippers and Tools

Parallel jaw grippers
Suction cups
Magnetic grippers
Custom end effectors
Tool changers



AI AND INTELLIGENCE

Machine Learning

Neural networks
Reinforcement learning
Deep learning
Transfer learning
Edge AI inference
Model optimization

Computer Vision

Object detection (YOLO, etc.)
Image segmentation
Pose estimation
Facial recognition
Feature tracking
Scene understanding

VLM Contextual

Vision-Language Models
Natural language commands
Scene description
Contextual reasoning
Multimodal understanding
LLM integration

Emergent Behavior

Swarm intelligence
Self-organization
Collective behavior
Multi-agent systems
Distributed intelligence
Flocking and formation

Navigation

Path planning (A*, RRT)
Obstacle avoidance
SLAM algorithms
Waypoint navigation
Dynamic re-planning
Map representation

Autonomous Systems

Decision making
Task planning
Behavior trees
State machines
Mission execution
Fail-safe systems



PLATFORMS AND TOOLS

Development Platforms

ROS (Robot Operating System)
Arduino ecosystem
Raspberry Pi
NVIDIA Jetson
ESP32/ESP8266

Simulation

Gazebo
Webots
CoppeliaSim (V-REP)
Unity robotics
PyBullet

Design Tools

CAD software (Fusion 360)
3D printing slicers
Circuit design (KiCAD)
FEA analysis
Kinematic modeling

Programming

Python for robotics
C++ for performance
Arduino C
MATLAB/Simulink
ROS nodes and packages

Libraries

OpenCV
TensorFlow/PyTorch
PCL (Point Cloud Library)
MoveIt (motion planning)
Navigation stack

Hardware Tools

3D printers
Soldering equipment
Oscilloscopes
Multimeters
Hand tools



LEARNING AND PROJECTS

Beginner Projects

Line following robot
Obstacle avoiding car
Arduino robot arm
RC car conversion
Simple gripper
LED matrix display

Intermediate Projects

Autonomous navigation
Computer vision tracking
Voice controlled robot
Hexapod walker
FPV drone build
SLAM mapping robot

Advanced Projects

Swarm robotics system
Bipedal walker
Autonomous delivery robot
AI-powered manipulation
Custom robot design
Competition robots

Learning Resources

Online Courses

Coursera, edX, Udacity robotics courses

Communities

Discord, Reddit, ROS community forums

Competitions

RoboCup, FIRST Robotics, combat robots

Books and Tutorials

Robotics textbooks, YouTube channels, blogs

Makerspaces

Shared workshops, equipment access, mentors

Guild Activities

Build sessions, demos, group projects



GUILD ACTIVITIES

Build Sessions

Hands-on robot building
Collaborative projects

Robot Demos

Show and tell
Live demonstrations

Competitions

Internal challenges
External contests

Workshops

Skills training
New technology demos

Study Groups

ROS tutorials
Algorithm discussions

Testing Days

Field testing
Performance tuning

Guest Speakers

Industry professionals
Research presentations

Troubleshooting

Debug sessions
Peer support

Social Events

Robot exhibitions
Networking meetups

Melbourne Tech Guilds • Robotics Guild

melbourne.techguilds.au